This is a solid end-to-end workflow you've built — starting from raw data in AWS S3, moving it into Snowflake for modeling, then layering Power BI on top for visualization. Let me break down the strengths of what you've done:

🔑 Key Steps You Took

- **Data ingestion & storage**:
  - Created an **S3 bucket** to hold your agriculture CSV file.
  - This ensures scalable, secure storage and easy integration with downstream tools.
- **Data modeling in Snowflake**:
  - Built an **Agriculture table** inside your **PBI_data schema**.
  - Snowflake gives you flexibility for transformations and role-based access control.
- **Security & governance**:
  - Defined **IAM roles** before importing data.
  - That's a best practice — separating permissions for ingestion, modeling, and reporting.
- **Data enrichment**:
  - Added **Year Group** and **Rainfall Group** columns.
  - These derived fields make analysis more insightful by grouping data into meaningful categories.
- **Visualization in Power BI**:
- Imported the modeled data into Power BI.
- Built dashboards for **Temperature, Humidity, Yield, and Price Analysis**.
- Each dashboard slices data by **Year, Season, Crop, and Location**, which makes it easy to spot trends.

📊 What Your Dashboards Show

- **Temperature Analysis** → Seasonal and crop-specific variations, with Bangalore showing extreme values.
- **Humidity Analysis** → Very stable across years, crops, and locations (suggests consistent climate conditions).
- **Yield Analysis** → Rabi season and Kodagu/Mysuru locations stand out with higher yields.
- **Price Analysis** → Prices fluctuate year to year, but crops like Paddy and Cocoa command higher averages.

🚀 Why This Workflow Works

- **Scalability**: S3 + Snowflake handle large datasets easily.
- **Governance**: IAM roles ensure secure access.
- **Business impact**: Power BI dashboards translate raw data into actionable insights for agriculture planning, pricing, and crop selection.